

Project #3

Enterprise Windows Infrastructure Administration Lab

Objective

Design, administer, maintain, troubleshoot, and document a Windows Server infrastructure that simulates the day-to-day responsibilities of a Junior Windows System Administrator.

This project focuses on **administration**, not initial deployment.

Technologies

Windows Server 2022

Active Directory

Group Policy

DNS

DHCP

VMware Workstation Pro

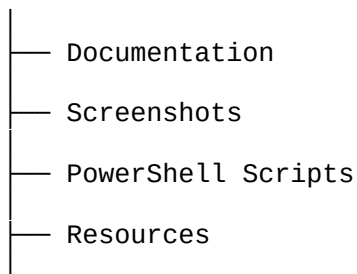
PowerShell

Windows 11

pfSense

Documentation Structure

Project 03 - Enterprise Windows Infrastructure Administration Lab



└─ ProjectReport.pdf
└─ README.md

Documentation Files

01-Project-Design.odt
02-Infrastructure-Overview.odt
03-Server-Roles.odt
04-PowerShell-Administration.odt
05-System-Administration-Tasks.odt
06-Troubleshooting.odt
07-Lessons-Learned.odt

Administration Modules

Module 1

Active Directory Administration

- User lifecycle
- Security groups
- OU delegation
- Fine-Grained Password Policy
- Restore deleted objects

Module 2

Group Policy Administration

- Administrative Templates
- Loopback Processing

- WMI Filters
- Security Filtering
- GPO Backup
- GPO Restore

Module 3

DNS Administration

- Forward Lookup Zones
- Reverse Lookup Zones
- Conditional Forwarders
- Aging & Scavenging
- Troubleshooting

Module 4

DHCP Administration

- Reservations
- Exclusions
- Scope Options
- Lease Management
- Troubleshooting

Module 5

File Server Administration

- DFS Namespace
- DFS Replication
- NTFS Permissions
- Shadow Copies
- File Screening
- Disk Quotas

Module 6

Windows Server Administration

- Windows Services
- Server Manager
- Event Viewer
- Performance Monitor
- Task Scheduler
- Windows Server Backup

Module 7

PowerShell Administration

- Bulk user creation
- Bulk password reset
- Export AD users
- Export group members
- Disable inactive accounts
- Generate reports

Module 8

Infrastructure Troubleshooting

- DNS failures
- DHCP failures
- Group Policy failures
- Authentication issues
- File share issues
- Replication issues

GitHub Repository

Enterprise-Windows-Infrastructure-Administration-Lab

Skills this project demonstrates

- Windows Server Administration
- Active Directory Administration
- Group Policy Management
- DNS Administration
- DHCP Administration
- PowerShell Administration
- File Server Administration
- Windows Infrastructure Troubleshooting
- VMware Administration
- Enterprise Documentation

What comes after Project #3

Once this is complete,

Project #4 – Wazuh SIEM & Endpoint Monitoring Lab